

June 9th, 2025

SWMS-Worksheet 1 Solutions: The Cantor Set

Consider $S_1 : [0, 1] \rightarrow \mathbb{R}$ and $S_2 : [0, 1] \rightarrow \mathbb{R}$, given by

$$\begin{aligned} S_1(x) &= \frac{1}{3}x, \quad \forall x \in \mathbb{R}, \\ S_2(x) &= \frac{1}{3}x + \frac{2}{3}, \quad \forall x \in \mathbb{R}. \end{aligned}$$

1. Using the above two *contracting affine maps* solve the following.

- (a) Show that both $S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)$ and $S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)$ are intervals.

Solution: We define

$$S_1[\alpha, \beta] = \{S_1(x) \mid x \in [\alpha, \beta]\} \text{ and } S_2[\alpha, \beta] = \{S_2(x) \mid x \in [\alpha, \beta]\}.$$

We will show that $S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) = \left[\frac{1}{9}, \frac{2}{9}\right]$ and $S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) = \left[\frac{7}{9}, \frac{8}{9}\right]$.

Let $x \in \left[\frac{1}{3}, \frac{2}{3}\right]$. Then $S_1(x) = \frac{x}{3} \in \left[\frac{1}{9}, \frac{2}{9}\right]$. Thus $S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) \subset \left[\frac{1}{9}, \frac{2}{9}\right]$.

Let $x \in \left[\frac{1}{9}, \frac{2}{9}\right]$. Then $3x \in \left[\frac{1}{3}, \frac{2}{3}\right]$ and $S_1(3x) = x$. Thus $x \in S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)$. This shows that $\left[\frac{1}{9}, \frac{2}{9}\right] \subset S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)$.

Since we have proved inclusion on both sides, we have $S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) = \left[\frac{1}{9}, \frac{2}{9}\right]$.

We will now show that $S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) = \left[\frac{7}{9}, \frac{8}{9}\right]$.

Let $x \in \left[\frac{1}{3}, \frac{2}{3}\right]$. Then $S_2(x) = \frac{x}{3} + \frac{2}{3} \in \left[\frac{7}{9}, \frac{8}{9}\right]$. Thus $S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) \subset \left[\frac{7}{9}, \frac{8}{9}\right]$.

Let $x \in \left[\frac{7}{9}, \frac{8}{9}\right]$. Then $3\left(x - \frac{2}{3}\right) \in \left[\frac{1}{3}, \frac{2}{3}\right]$ and $S_2(3x - 2) = x$. Thus $x \in S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)$. This shows that $\left[\frac{7}{9}, \frac{8}{9}\right] \subset S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)$.

Since we have proved inclusion on both sides, we have $S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) = \left[\frac{7}{9}, \frac{8}{9}\right]$.

- (b) Plot them and observe that $S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) \cup S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)$ is a union of two disjoint intervals.

Solution:



- (c) What is the length of each of the intervals $S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)$ and $S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)$?

Solution:

$$S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) = \left[\frac{1}{9}, \frac{2}{9}\right] \implies \text{diam}\left(S_1\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)\right) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9}.$$

$$S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right) = \left[\frac{7}{9}, \frac{8}{9}\right] \implies \text{diam}\left(S_2\left(\left[\frac{1}{3}, \frac{2}{3}\right]\right)\right) = \frac{8}{9} - \frac{7}{9} = \frac{1}{9}.$$

- (d) Repeat the above for the interval $[\alpha, \beta]$ where $0 \leq \alpha \leq \beta \leq 1$.

Solution:

Repeating the solutions for the above three parts for $[\alpha, \beta]$, we get that $S_1([\alpha, \beta]) = \left[\frac{\alpha}{3}, \frac{\beta}{3}\right]$ and $S_2([\alpha, \beta]) = \left[\frac{\alpha}{3} + \frac{2}{3}, \frac{\beta}{3} + \frac{2}{3}\right]$.

The intervals will be disjoint $\iff \frac{\alpha}{3} + \frac{2}{3} > \frac{\beta}{3} \iff \beta - \alpha < 2$, which is true $\forall 0 \leq \alpha \leq \beta \leq 1$. Thus the intervals will always be disjoint.

The length of the intervals $S_1([\alpha, \beta])$ and $S_2([\alpha, \beta])$ is $\frac{\beta - \alpha}{3} = \frac{\text{diam}([\alpha, \beta])}{3}$.

2. Define C_0, C_1, C_2 iteratively, as follows:

$$\begin{aligned} C_0 &= [0, 1], \\ C_1 &= S_1(C_0) \cup S_2(C_0), \\ C_2 &= S_1(C_1) \cup S_2(C_1). \end{aligned}$$

- (a) Find C_0, C_1 , and C_2 .

Solution: $C_0 = [0, 1]$ by definition.

We have shown in 1d that

$$S_1([\alpha, \beta]) = \left[\frac{\alpha}{3}, \frac{\beta}{3}\right]$$

and

$$S_2([\alpha, \beta]) = \left[\frac{\alpha}{3} + \frac{2}{3}, \frac{\beta}{3} + \frac{2}{3}\right].$$

Thus

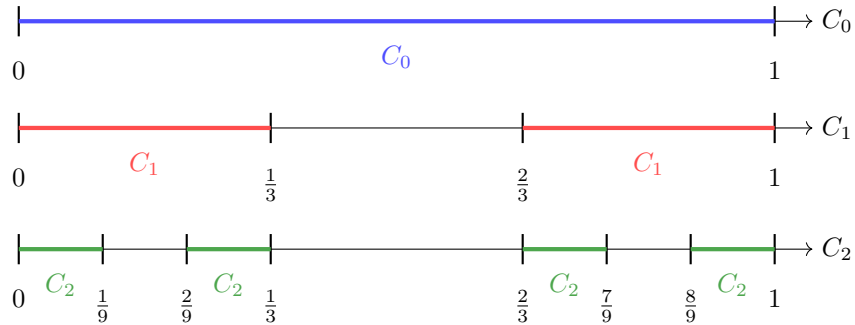
$$C_1 = S_1([0, 1]) \cup S_2([0, 1]) = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right].$$

Now,

$$\begin{aligned} C_2 &= S_1(C_1) \cup S_2(C_1) = S_1\left(\left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]\right) \cup S_2\left(\left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]\right) \\ &= S_1\left(\left[0, \frac{1}{3}\right]\right) \cup S_1\left(\left[\frac{2}{3}, 1\right]\right) \cup S_2\left(\left[0, \frac{1}{3}\right]\right) \cup S_2\left(\left[\frac{2}{3}, 1\right]\right) = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{8}{9}, 1\right]. \end{aligned}$$

- (b) Plot C_0, C_1, C_2 in separate graphs, inside the interval $[0, 1]$.

Plots:



- (c) For $n = 0, 1, 2$, let N_n be the number of disjoint intervals and ℓ_n be the length of any one of the intervals in C_n (respectively). Fill in the table below:

n	N_n	ℓ_n
0	1	1
1	2	$\frac{1}{3}$
2	4	$\frac{1}{9}$

3. For each $n \geq 1$, let $C_n = S_1(C_{n-1}) \cup S_2(C_{n-1})$.

- (a) Show that

$$N_{n+1} = 2N_n, \quad \ell_{n+1} = \frac{\ell_n}{3}, \quad \forall n \geq 1.$$

What is the significance of the scaling constants 2 and $\frac{1}{3}$ in the above relations?

Solution:

We have shown in 1d that $\forall 0 \leq \alpha \leq \beta \leq 1$, the images $S_1([\alpha, \beta])$ and $S_2([\alpha, \beta])$ are disjoint intervals.

Say C_n has N_n disjoint intervals, each of length ℓ_n . Then,

$$C_n = \bigsqcup_{i=1}^{N_n} [a_i, b_i] \quad \text{such that } b_i - a_i = \ell_n \quad \forall i \in \{1, 2, \dots, N_n\}.$$

Then,

$$C_{n+1} = S_1(C_n) \cup S_2(C_n) = \bigsqcup_{i=1}^{N_n} \left[\frac{a_i}{3}, \frac{b_i}{3} \right] \cup \left[\frac{a_i + 2}{3}, \frac{b_i + 2}{3} \right].$$

Since $\{[a_i, b_i]\}_{i=1}^{N_n}$ were disjoint, the images

$$\left\{ \left[\frac{a_i}{3}, \frac{b_i}{3} \right] \right\}_{i=1}^{N_n}$$

are disjoint and the images

$$\left\{ \left[\frac{a_i + 2}{3}, \frac{b_i + 2}{3} \right] \right\}_{i=1}^{N_n}$$

are also disjoint. Also,

$$\left\{ \left[\frac{a_i}{3}, \frac{b_i}{3} \right], \left[\frac{a_i + 2}{3}, \frac{b_i + 2}{3} \right] \right\}_{i=1}^{N_n}$$

are disjoint since S_1 and S_2 map disjoint intervals in $[0, 1]$ to disjoint intervals.

Thus, C_{n+1} is a disjoint union of $N_{n+1} = 2N_n$ intervals, each of length $\frac{b_i - a_i}{3} = \frac{\ell_n}{3} = \ell_{n+1}$.

Conclusion: Thus, at each iteration, the middle-third segment of each interval in C_n is removed, and the remaining two intervals are of one-third the original length.

(b) Show that C_n is a proper subset of C_{n-1} , for all $n \geq 1$.

Solution: We will proceed by induction on n . We know that:

$$C_0 = [0, 1] \Rightarrow C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right] \subset C_0.$$

We assume $C_n \subset C_{n-1}$ for some $n \geq 1$.

Then:

$$C_{n+1} = S_1(C_n) \cup S_2(C_n).$$

Let $x \in C_{n+1}$.

Then:

$$x \in S_1(C_n) \quad \text{or} \quad x \in S_2(C_n) \Leftrightarrow \exists y \in C_n \text{ such that } x = \frac{y}{3} \text{ or } x = \frac{y}{3} + \frac{2}{3}.$$

Since $y \in C_n \subset C_{n-1}$, we have:

$$S_1(y) \in S_1(C_n) \subset S_1(C_{n-1}), \quad S_2(y) \in S_2(C_n) \subset S_2(C_{n-1}).$$

Thus,

$$S_1(y), S_2(y) \in S_1(C_{n-1}) \cup S_2(C_{n-1}) = C_n.$$

If $x \in S_1(C_n)$, then $x = S_1(y) \in C_n$.

If $x \in S_2(C_n)$, then $x = S_2(y) \in C_n$.

Thus $x \in C_{n+1} \Rightarrow x \in C_n \Rightarrow C_{n+1} \subset C_n$ and by the principle of induction, we are done. Now we will show that the inclusion is proper. We have shown in [3a](#) that the length of the intervals changes as $l_{n+1} = \frac{l_n}{3}$ where $l_0 = \text{diam}([0, 1]) = 1$. This gives us the expression $l_n = 3^{-n}$.

We have also shown in [3a](#) that the number of disjoint intervals changes as $N_{n+1} = 2N_n$ where N_0 is the number of disjoint intervals in $C_0 = [0, 1] \implies N_0 = 1$. This gives us the expression $N_n = 2^n$. Thus C_n is the union of 2^n disjoint intervals of length 3^{-n} each. Then the total length of the intervals of $C_n = L_n = N_n l_n = \left(\frac{2}{3}\right)^n$, which is strictly decreasing in $n \in \mathbb{N}$.

Thus

$$L_{n-1} - L_n = \left(\frac{2}{3}\right)^{n-1} - \left(\frac{2}{3}\right)^n = \frac{1}{3} \cdot \left(\frac{2}{3}\right)^{n-1} > 0 \quad \forall n \in \mathbb{N}.$$

If $C_{n-1} \setminus C_n = \emptyset$ for some $n \in \mathbb{N}$, $L_{n-1} - L_n = 0$, which is not true. Hence

$$C_{n-1} \setminus C_n \neq \emptyset \quad \forall n \in \mathbb{N}.$$

(c) Can you find 5 points in $\bigcap_{n=1}^{\infty} C_n$?

Solution:

Claim:

$$\left\{0, \frac{1}{4}, \frac{1}{3}, \frac{3}{4}, 1\right\} \subseteq \bigcap_{n=1}^{\infty} C_n.$$

We will proceed by induction on n . We know that:

$$\left\{0, \frac{1}{4}, \frac{1}{3}, \frac{3}{4}, 1\right\} \subseteq [0, 1] = C_0.$$

Let

$$\left\{0, \frac{1}{4}, \frac{1}{3}, \frac{3}{4}, 1\right\} \subseteq C_n \text{ for some } n \in \mathbb{N}.$$

Then:

$$C_{n+1} = S_1(C_n) \cup S_2(C_n) = \left\{\frac{x}{3} \mid x \in C_n\right\} \cup \left\{\frac{x+2}{3} \mid x \in C_n\right\}.$$

- $0 \in C_n \Rightarrow \frac{0}{3} = 0 \in C_{n+1}.$
- $1 \in C_n \Rightarrow \frac{1+2}{3} = 1 \in C_{n+1}.$
- $1 \in C_n \Rightarrow \frac{1}{3} \in C_{n+1}.$
- $\frac{3}{4} \in C_n \Rightarrow \frac{\left(\frac{3}{4}\right)}{3} = \frac{1}{4} \in C_{n+1}.$
- $\frac{1}{4} \in C_n \Rightarrow \frac{\left(\frac{1}{4}+2\right)}{3} = \frac{3}{4} \in C_{n+1}.$

Thus:

$$\left\{0, \frac{1}{4}, \frac{1}{3}, \frac{3}{4}, 1\right\} \subseteq C_{n+1}.$$

By the principle of induction:

$$\left\{0, \frac{1}{4}, \frac{1}{3}, \frac{3}{4}, 1\right\} \subseteq \bigcap_{n=1}^{\infty} C_n.$$

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SWMS-Worksheet 2 Solutions: The Sierpinski Triangle

Let Δ be an equilateral triangle with vertices $P = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $Q = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $R = \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}$. Let T be the region enclosing and including the triangle Δ . Consider $S_1, S_2, S_3 : T \rightarrow \mathbb{R}^2$ given by

$$\begin{aligned} S_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{2} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2, \\ S_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{2} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2, \\ S_3 \begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{2} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix}, \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2. \end{aligned}$$

1. Plot T with labels P , Q and R for the vertices and for the three *contracting affine maps* on T solve the following:

- (a) Show that if $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$ is any point in T , then

$$S_1(\mathbf{x}) = \frac{\mathbf{x} + P}{2}, \quad S_2(\mathbf{x}) = \frac{\mathbf{x} + Q}{2}, \quad S_3(\mathbf{x}) = \frac{\mathbf{x} + R}{2}.$$

Explain the action of the maps geometrically on T .

Solution: Let $\vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}$. Then:

$$\begin{aligned} S_1(\vec{x}) &= \frac{1}{2} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix}}{2} = \frac{\vec{x} + P}{2}, \\ S_2(\vec{x}) &= \frac{1}{2} \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}}{2} = \frac{\vec{x} + Q}{2}, \\ S_3(\vec{x}) &= \frac{1}{2} \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix} = \frac{\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}}{2} = \frac{\vec{x} + R}{2}. \end{aligned}$$

This tells us that $S_1(x)$ maps x to the midpoint on P and x , $S_2(x)$ maps x to the midpoint of Q and x and $S_3(x)$ maps x to the midpoint of R and x .

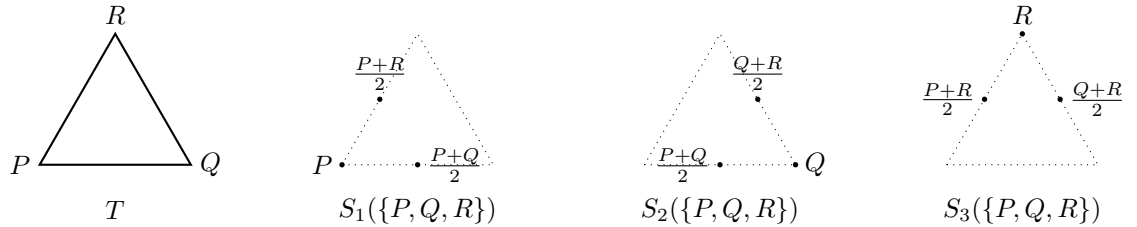
- (b) In separate figures, plot the images of the vertices P, Q, R of the triangle Δ under the three maps S_1, S_2 , and S_3 .

Solution: We have

$$S_1(P) = P, \quad S_1(Q) = \frac{P+Q}{2} \quad \text{and} \quad S_1(R) = \frac{P+R}{2},$$

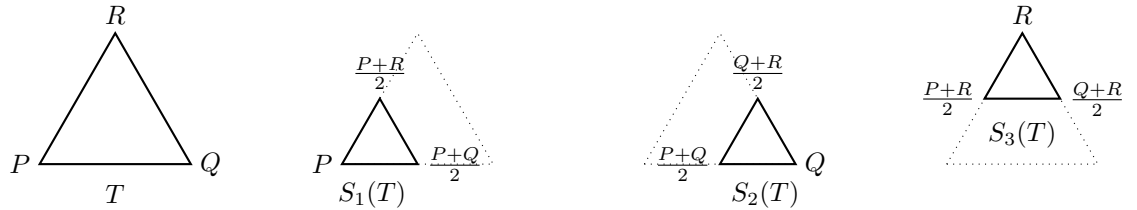
$$S_2(P) = \frac{P+Q}{2}, \quad S_2(Q) = Q \quad \text{and} \quad S_2(R) = \frac{Q+R}{2} \quad \text{and}$$

$$S_3(P) = \frac{P+R}{2}, \quad S_3(Q) = \frac{Q+R}{2} \quad \text{and} \quad S_3(R) = R.$$



- (c) Plot the image of the triangle Δ under the three maps S_1, S_2 , and S_3 . *Hint: Q3 Night homework June 6th, 2025*

Solution: Let us consider triangle T with vertices P, Q, R , and its images under maps S_1, S_2, S_3 :



- (d) Show that each of $S_1(T), S_2(T), S_3(T)$ are triangular regions properly contained in T . What are their areas? Can you find a point in T which is not in $S_1(T) \cup S_2(T) \cup S_3(T)$?

Solution: Let us consider the line segments $\mathcal{L}(P, Q)$, $\mathcal{L}(Q, R)$ and $\mathcal{L}(R, P)$, the sides of T . As proved in night homework, S_1, S_2 and S_3 are affine maps and hence map line segments to line segments. Thus:

- $S_1(\mathcal{L}(P, Q)) = \mathcal{L}\left(P, \frac{P+Q}{2}\right), S_1(\mathcal{L}(Q, R)) = \mathcal{L}\left(\frac{P+Q}{2}, \frac{P+R}{2}\right), S_1(\mathcal{L}(R, P)) = \mathcal{L}\left(\frac{P+R}{2}, P\right).$
- $S_2(\mathcal{L}(P, Q)) = \mathcal{L}\left(\frac{P+Q}{2}, Q\right), S_2(\mathcal{L}(Q, R)) = \mathcal{L}\left(Q, \frac{Q+R}{2}\right), S_2(\mathcal{L}(R, P)) = \mathcal{L}\left(\frac{Q+R}{2}, \frac{P+Q}{2}\right).$
- $S_3(\mathcal{L}(P, Q)) = \mathcal{L}\left(\frac{P+R}{2}, \frac{Q+R}{2}\right), S_3(\mathcal{L}(Q, R)) = \mathcal{L}\left(\frac{Q+R}{2}, R\right), S_3(\mathcal{L}(R, P)) = \mathcal{L}\left(R, \frac{P+R}{2}\right).$

We see that each of S_1, S_2 and S_3 map the sides of a larger equilateral triangle T to the sides of a smaller equilateral triangle of half the length. Let us now consider an internal point $x \in T$, one that may not lie on its boundary. Since T is a triangle, x lies on a line segment joining a vertex $y \in \{P, Q, R\}$ to a point $z \in T$ that lies on one of the edges of T . Thus $x \in \mathcal{L}(y, z)$. We know that S_1, S_2 and S_3 map $\mathcal{L}(y, z)$ to a line segment such that

$$S_i(\mathcal{L}(y, z)) = \mathcal{L}(S_i(y), S_i(z)) \quad \forall 1 \leq i \leq 3.$$

Since $x \in \mathcal{L}(y, z)$, $\forall 1 \leq i \leq 3$,

$$S_i(x) \in S_i(\mathcal{L}(y, z)) = \mathcal{L}(S_i(y), S_i(z)) \in T.$$

This shows that S_1, S_2 and S_3 map the entire solid triangle T to smaller solid triangles contained in T .

$S_1(T)$, $S_2(T)$ and $S_3(T)$ each have a side-length of $\frac{\|P-Q\|}{2} = \frac{1}{2}$. Since they are equilateral triangles, each of them have an area of $\frac{\sqrt{3}}{4} \left(\frac{1}{2}\right)^2 = \frac{\sqrt{3}}{16}$.

Claim:

$$\left(\frac{1}{2}, \frac{1}{4}\right) \notin S_1(T) \cup S_2(T) \cup S_3(T).$$

$$\left(\frac{1}{2}, \frac{1}{4}\right) \in S_1(T) \iff \exists \vec{x} \in T \text{ such that } S_1(\vec{x}) = \left(\frac{1}{2}, \frac{1}{4}\right)$$

$$\iff \frac{1}{2}\vec{x} + \frac{1}{2}P = \left(\frac{1}{2}, \frac{1}{4}\right) \iff \vec{x} = \left(1, \frac{1}{2}\right) \notin T$$

which is a contradiction.

$$\left(\frac{1}{2}, \frac{1}{4}\right) \in S_2(T) \iff \exists \vec{x} \in T \text{ such that } S_2(\vec{x}) = \left(\frac{1}{2}, \frac{1}{4}\right)$$

$$\iff \frac{1}{2}\vec{x} + \frac{1}{2}Q = \left(\frac{1}{2}, \frac{1}{4}\right) \iff \vec{x} = \left(0, \frac{1}{2}\right) \notin T$$

which is a contradiction.

$$\left(\frac{1}{2}, \frac{1}{4}\right) \in S_3(T) \iff \exists \vec{x} \in T \text{ such that } S_3(\vec{x}) = \left(\frac{1}{2}, \frac{1}{4}\right)$$

$$\iff \frac{1}{2}\vec{x} + \frac{1}{2}R = \left(\frac{1}{2}, \frac{1}{4}\right) \iff \vec{x} = \left(\frac{1}{2}, \frac{1-\sqrt{3}}{2}\right) \notin T$$

which is a contradiction.

Thus,

$$\left(\frac{1}{2}, \frac{1}{4}\right) \notin S_1(T) \cup S_2(T) \cup S_3(T).$$

This also shows that the inclusion $S_i(T) \subset T \quad \forall 1 \leq i \leq 3$ is proper.

2. Define T_0, T_1, T_2 iteratively, as follows:

$$\begin{aligned} T_0 &= T, \\ T_1 &= S_1(T_0) \cup S_2(T_0) \cup S_3(T_0), \\ T_2 &= S_1(T_1) \cup S_2(T_1) \cup S_3(T_1). \end{aligned}$$

(a) Find T_0, T_1 , and T_2 . Plot T_0, T_1 , and T_2 , in separate graphs.

Solution: By definition, $T_0 = T$.

Also, as shown in 1d,

- $S_1(T)$ is the solid triangle with $S_1(P) = P, S_1(Q) = \frac{P+Q}{2}$ and $S_1(R) = \frac{P+R}{2}$ as its vertices.
- $S_2(T)$ is the solid triangle with $S_2(P) = \frac{P+Q}{2}, S_2(Q) = Q$ and $S_2(R) = \frac{Q+R}{2}$ as its vertices.
- $S_3(T)$ is the solid triangle with $S_3(P) = \frac{P+R}{2}, S_3(Q) = \frac{Q+R}{2}$ and $S_3(R) = R$ as its vertices.

Thus $T_1 = S_1(T) \cup S_2(T) \cup S_3(T) \subset T_0$ is a set of the above three equilateral triangles sharing one common vertex each.

We now compute T_2 using T_1 using the formula:

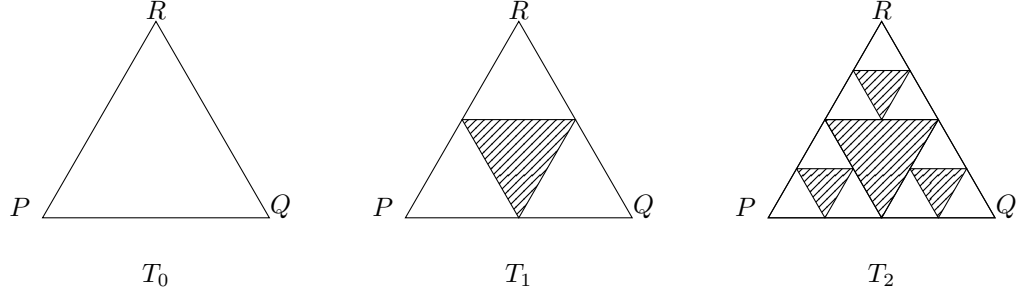
$$T_2 = \bigcup_{1 \leq i, j \leq 3} S_i(S_j(T)).$$

Again, we map the vertices P, Q and R of T to form the images under $S_i \circ S_j$. We have,

- $S_1(S_1(P)) = (0, 0), \quad S_1(S_1(Q)) = \left(\frac{1}{4}, 0\right), \quad S_1(S_1(R)) = \left(\frac{1}{4}, \frac{\sqrt{3}}{4}\right)$
- $S_1(S_2(P)) = \left(\frac{1}{4}, 0\right), \quad S_1(S_2(Q)) = \left(\frac{1}{2}, 0\right), \quad S_1(S_2(R)) = \left(\frac{3}{8}, \frac{\sqrt{3}}{4}\right)$
- $S_1(S_3(P)) = \left(\frac{1}{8}, \frac{\sqrt{3}}{8}\right), \quad S_1(S_3(Q)) = \left(\frac{3}{8}, \frac{\sqrt{3}}{8}\right), \quad S_1(S_3(R)) = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}\right)$
- $S_2(S_1(P)) = \left(\frac{1}{2}, 0\right), \quad S_2(S_1(Q)) = \left(\frac{3}{4}, 0\right), \quad S_2(S_1(R)) = \left(\frac{3}{4}, \frac{\sqrt{3}}{4}\right)$
- $S_2(S_2(P)) = \left(\frac{3}{4}, 0\right), \quad S_2(S_2(Q)) = (1, 0), \quad S_2(S_2(R)) = \left(\frac{7}{8}, \frac{\sqrt{3}}{4}\right)$
- $S_2(S_3(P)) = \left(\frac{5}{8}, \frac{\sqrt{3}}{8}\right), \quad S_2(S_3(Q)) = \left(\frac{7}{8}, \frac{\sqrt{3}}{8}\right), \quad S_2(S_3(R)) = \left(1, \frac{\sqrt{3}}{4}\right)$
- $S_3(S_1(P)) = \left(\frac{1}{4}, \frac{\sqrt{3}}{4}\right), \quad S_3(S_1(Q)) = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}\right), \quad S_3(S_1(R)) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
- $S_3(S_2(P)) = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}\right), \quad S_3(S_2(Q)) = \left(\frac{3}{4}, \frac{\sqrt{3}}{4}\right), \quad S_3(S_2(R)) = \left(\frac{7}{8}, \frac{\sqrt{3}}{2}\right)$
- $S_3(S_3(P)) = \left(\frac{3}{8}, \frac{3\sqrt{3}}{8}\right), \quad S_3(S_3(Q)) = \left(\frac{5}{8}, \frac{3\sqrt{3}}{8}\right), \quad S_3(S_3(R)) = \left(\frac{3}{4}, \frac{3\sqrt{3}}{4}\right)$

$\forall 1 \leq i, j \leq 3$, $S_i(S_j(P)), S_i(S_j(Q))$ and $S_i(S_j(R))$ give us the vertices of a triangle in T_2 and by definition of T_2 , the 9 solid triangles constitute T_2 .

Plot:



- (b) For $n = 0, 1, 2$, let N_n be the number of disjoint triangular regions and A_n be the area of any triangular region in T_n . Fill in the table below:

n	N_n	A_n
0	1	$\sqrt{3}/4$
1	3	$\sqrt{3}/16$
2	9	$\sqrt{3}/64$

3. For $n \geq 1$, let $T_n = S_1(T_{n-1}) \cup S_2(T_{n-1}) \cup S_3(T_{n-1})$.

- (a) Show that

$$N_{n+1} = 3N_n, \quad A_{n+1} = \frac{A_n}{4}, \quad \forall n \geq 1.$$

Solution:

Before starting our proof, we prove that S_1, S_2 and S_3 are one-one maps. Let for some $x, y \in T$, $S_1(x) = S_1(y)$. Then $S_1(x) = S_1(y) \iff \frac{x+P}{2} = \frac{y+P}{2} \iff x = y$. We repeat this proof for S_2 and S_3 as $S_2(x) = \frac{x+Q}{2}$ and $S_3(x) = \frac{x+R}{2}$. Thus we are done.

We assume that the triangles in T_n sharing at most one common point are all equilateral with equal side length. Let the number of such triangles in T_n be N_n with area A_n .

We have $T_{n+1} = S_1(T_n) \cup S_2(T_n) \cup S_3(T_n)$. Since we know that each of the maps S_1, S_2 and S_3 map T_n into three affine copies of T_n inside $S_1(T_0), S_2(T_0)$ and $S_3(T_0)$ respectively, they remain disjoint or just connected at the vertices (refer to Figure 1c). Also since S_1, S_2 and S_3 are one-one, they have unique images and hence if two triangles of T_n are disjoint or connected at the vertex only, their images under each map will also share at most one common point. We will prove the above explanation rigorously below.

Let the N_n triangles in T_n be denoted by $\{U_i\}_{i=1}^{N_n}$. Since S'_k s are one-one $\forall k \in \{1, 2, 3\}$, we have that

$$\forall i \neq j \in \{1, 2, \dots, N_n\}, \quad |U_i \cap U_j| \leq 1 \implies |S_k(U_i) \cap S_k(U_j)| \leq 1 \quad \forall k \in \{1, 2, 3\}.$$

We also use the fact that $\forall i$, $S_1(U_i) \subset S_1(T_0)$, $S_2(U_i) \subset S_2(T_0)$ and $S_3(U_i) \subset S_3(T_0)$. Since $S_1(T_0), S_2(T_0)$ and $S_3(T_0)$ have at most one point of intersection, $\forall k \neq l \in$

$\{1, 2, 3\}$, we have

$$\forall i \neq j \in \{1, 2, \dots, N_n\}, |S_k(U_i) \cap S_l(U_j)| \leq 1.$$

The two statements above shows that $\forall k, l \in \{1, 2, 3\}$, we have

$$\forall i \neq j \in \{1, 2, \dots, N_n\}, |S_k(U_i) \cap S_l(U_j)| \leq 1.$$

Thus

$$N_{n+1} = |\{S_k(U_i) : k \in \{1, 2, 3\}, i \in \{1, 2, \dots, N_n\}\}| = 3N_n.$$

Let the vertices of U_i be given by P_i, Q_i, R_i . Since S_1, S_2 and S_3 are affine contractions, $S_1(U_i)$ will have vertices $\frac{P_i+P}{2}, \frac{Q_i+P}{2}$ and $\frac{R_i+P}{2}$. Similarly, $S_2(U_i)$ will have vertices $\frac{P_i+Q}{2}, \frac{Q_i+Q}{2}$ and $\frac{R_i+Q}{2}$ and $S_3(U_i)$ will have vertices $\frac{P_i+R}{2}, \frac{Q_i+R}{2}$ and $\frac{R_i+R}{2}$. By our assumption,

$$\forall i, j, k \in \{1, 2, \dots, N_n\}, \|P_i - Q_i\| = \|Q_j - R_j\| = \|R_k - P_k\|.$$

It is easy to see that the triangles of T_{n+1} are also equilateral with side lengths of

$$\left\| \frac{P_1 - Q_1}{2} \right\| = \frac{1}{2} \|P_1 - Q_1\|.$$

Thus T_{n+1} has $N_{n+1} = 3N_n$ triangles of area $A_{n+1} = \left(\frac{1}{2}\right)^2 A_n = \frac{A_n}{4}$ and we are done.

(b) Show that T_n is a proper subset of T_{n-1} , for all $n \geq 1$.

Solution: We have shown in 2a that $T_0 \supset T_1$. Now we will proceed by induction on n . Let $T_n \subset T_{n-1}$ for some $n \geq 0$. Then

$$S_1(T_n) \subset S_1(T_{n-1}), S_2(T_n) \subset S_2(T_{n-1}) \text{ and } S_3(T_n) \subset S_3(T_{n-1}).$$

Taking union,

$$\begin{aligned} S_1(T_n) \cup S_2(T_n) \cup S_3(T_n) &\subset S_1(T_{n-1}) \cup S_2(T_{n-1}) \cup S_3(T_{n-1}) \\ \implies T_{n+1} &\subset T_n \end{aligned}$$

by definition and we are done.

Now, at the n^{th} step, we have $N_n = 3^n$ triangles of area $A_n = \frac{\sqrt{3}}{4} \cdot 4^{-n}$ each. Thus the total area covered by T_n is $N_n \cdot A_n = \frac{\sqrt{3}}{4} \cdot \left(\frac{3}{4}\right)^n$ which is strictly decreasing in $n \in \mathbb{N}$. Thus $\forall n \geq 1$, the difference in the area covered by T_{n-1} and T_n is

$$N_{n-1} \cdot A_{n-1} - N_n \cdot A_n = \frac{\sqrt{3}}{4} \cdot \left(\frac{3}{4}\right)^{n-1} - \frac{\sqrt{3}}{4} \cdot \left(\frac{3}{4}\right)^n = \frac{\sqrt{3}}{16} \cdot \left(\frac{3}{4}\right)^{n-1} > 0.$$

If for some $n \in \mathbb{N}$, $T_{n-1} \setminus T_n = \emptyset$, area covered by $T_{n-1} \setminus T_n = 0$. Thus

$$T_{n-1} \setminus T_n \neq \emptyset \forall n \geq 1.$$

June 9th, 2025

SWMS-Worksheet 3 Solutions: Construction of Cantor Set and Sierpinski Triangle

1. Recall $C_0, C_1, C_2, C_3, \dots, C_n, \dots$ from Worksheet 1. Define the set $\mathcal{C}$, known as the *middle-thirds Cantor Set*, by

$$\mathcal{C} = \bigcap_{n=0}^{\infty} C_n.$$

- (a) Decide if the following are true or false (with justification):

- i. It is possible that $x \notin C_{100}$, but $x \in \mathcal{C}$?

Solution: Since $\mathcal{C} = \bigcap_{n=0}^{\infty} C_n$,

$$x \in \mathcal{C} \iff x \in C_n \forall n \in \mathbb{N} \implies x \in C_{100}.$$

Thus

$$x \notin C_{100} \implies x \notin \mathcal{C} \text{ and thus } \mathcal{C} \setminus C_{100} = \emptyset.$$

Thus $\nexists$ such an x .

- ii. $\mathcal{C}$ is the largest set that is contained in each C_n for all $n \geq 1$.

Solution: Let $\mathcal{C} = \bigcap_{n=0}^{\infty} C_n$. Let $D \subset \mathbb{R}$ such that $D \subseteq C_n \forall n \geq 1$.

Then

$$x \in D \implies x \in C_n \forall n \geq 1 \implies x \in \mathcal{C}.$$

Thus $D \subseteq \mathcal{C}$. Also, by definition,

$$\bigcap_{n=1}^{\infty} C_n \subseteq C_n \quad \forall n \geq 1.$$

Thus $\mathcal{C}$ is largest set contained in each C_n .

- (b) Show that both 0 and 1 belong to $\mathcal{C}$.

Solution: Shown in 3c.

- (c) Which of the following points lie in $\mathcal{C}$?

$$1/3, 2/3, 1/2, 1/4, 1/9.$$

Solution: Claim: $\{1/3, 2/3, 1/4, 1/9\} \in \mathcal{C}$ and $\frac{1}{2} \notin \mathcal{C}$.

- i. We have already shown in 3c that $\{1/3, 1/4\} \in \mathcal{C}$.

- ii. $C_1 = [0, 1/3] \cup [2/3, 1] \implies 1/2 \notin C_1$ and since

$$C_n \subset C_{n-1} \subset C_1 \forall n \geq 1, \quad 1/2 \notin C_n \forall n \geq 1.$$

Thus $1/2 \notin \mathcal{C}$.

iii.

$$0 \in C_n \implies S_2(0) = \frac{0}{3} + \frac{2}{3} = \frac{2}{3} \in C_{n+1}.$$

Since $0 \in \mathcal{C}$, $0 \in C_n \forall n \geq 0$. Thus $\frac{2}{3} \in C_n \forall n \geq 1$. We also know $\frac{2}{3} \in [0, 1] = C_0$.
Thus we have

$$\frac{2}{3} \in C_n \forall n \geq 0 \implies \frac{2}{3} \in \mathcal{C}.$$

iv. We repeat the argument above as $\frac{1}{3} \in \mathcal{C}$ and $\frac{1}{9} = S_1(\frac{1}{3})$.

(d) What is the total of the lengths of all the intervals in C_0, C_1, C_2 ? What is it for C_n ?

Solution: Shown in 3b.

(e) Does the total of the lengths converge as $n \rightarrow \infty$? If yes, what is the limit?

Solution: We have shown in 3b that the total length of the intervals of $C_n = L_n = N_n l_n = \left(\frac{2}{3}\right)^n$.

Claim: $\forall 0 < a < 1, a^n \rightarrow 0$ as $n \rightarrow \infty$.

Let $\epsilon > 0$ be arbitrary.

$$\because 0 < a < 1, \exists 0 < \delta < 1 \text{ such that } a = \frac{1}{1+\delta}.$$

Then

$$a^n = \frac{1}{(1+\delta)^n} = \frac{1}{1+n\delta + \dots + \binom{n}{n}\delta^n} < \frac{1}{n\delta}.$$

We define $N = \frac{1}{\epsilon\delta}$. Then

$$n > N \implies a^n = |a^n - 0| < \frac{1}{n\delta} < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, by definition of convergence we have proved our claim.

Here, $L_n = \left(\frac{2}{3}\right)^n$. Since $0 < \frac{2}{3} < 1$, using the lemma above we have $L_n \rightarrow 0$ as $n \rightarrow \infty$.

(f) Show that $\mathcal{C} = S_1(\mathcal{C}) \cup S_2(\mathcal{C})$.

Solution: Let $x \in \mathcal{C} = \bigcap_{n=0}^{\infty} C_n$. Thus $x \in C_n \forall n \geq 0$.

• Case 1:

$$x \in \left[0, \frac{1}{3}\right].$$

$S_1(\mathcal{C}) = \{\frac{a}{3} \mid a \in \mathcal{C}\}$. Let $y=3x$.

If $y \in \mathcal{C}$, $x = \frac{y}{3} \in S_1(\mathcal{C})$.

If $y \notin \mathcal{C}$, $\exists n \in \mathbb{N}$ such that $y \notin C_n$. But $x \in C_{n+1}$ as $x \in \mathcal{C}$.

Since $x \in [0, \frac{1}{3}]$, $x \in S_1(C_n) = \left\{\frac{a}{3} \mid a \in C_n\right\}$. Then $\exists a \in C_n$ such that $x = \frac{a}{3} \implies$

$a = 3x = y$. This contradicts $y \notin C_n$.

Thus $y \in \mathcal{C}$ and we are done.

• Case 2:

$$x \in \left[\frac{2}{3}, 1\right].$$

$S_2(\mathcal{C}) = \{\frac{a}{3} + \frac{2}{3} \mid a \in \mathcal{C}\}$. Let $y=3x-2$.

If $y \in \mathcal{C}$, $x = \frac{y}{3} + \frac{2}{3} \in S_2(\mathcal{C})$.

If $y \notin \mathcal{C}$, $\exists n \in \mathbb{N}$ such that $y \notin C_n$. But $x \in C_{n+1}$ as $x \in \mathcal{C}$.

Since $x \in [\frac{2}{3}, 1]$, $x \in S_2(C_n) = \left\{ \frac{a}{3} + \frac{2}{3} \mid a \in C_n \right\}$. Then $\exists a \in C_n$ such that

$$x = \frac{a}{3} + \frac{2}{3} \implies a = 3x - 2 = y. \text{ This contradicts } y \notin C_n.$$

Thus $y \in \mathcal{C}$ and we are done.

Since x must belong to $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, we conclude:

$$\mathcal{C} \subseteq S_1(\mathcal{C}) \cup S_2(\mathcal{C}).$$

Let $x \in S_1(\mathcal{C}) \cup S_2(\mathcal{C})$.

- Case 1:

$$x \in S_1(\mathcal{C}).$$

Then $x = \frac{y}{3}$ for some $y \in \mathcal{C}$. Now, $y \in C_n \implies x = \frac{y}{3} \in C_{n+1}$. Since $y \in \mathcal{C}$, $y \in C_n \forall n \geq 0$. Thus $x \in C_n \forall n \geq 1$. Since $C_0 \supset C_1$, $x \in C_n \forall n \geq 0$. Thus, $x \in \mathcal{C}$.

- Case 2:

$$x \in S_2(\mathcal{C}).$$

Then $x = \frac{y}{3} + \frac{2}{3}$ for some $y \in \mathcal{C}$. Now, $y \in C_n \implies x = \frac{y}{3} + \frac{2}{3} \in C_{n+1}$. Since $y \in \mathcal{C}$, $y \in C_n \forall n \geq 0$. Thus $x \in C_n \forall n \geq 1$. Since $C_0 \supset C_1$, $x \in C_n \forall n \geq 0$. Thus, $x \in \mathcal{C}$.

Therefore:

$$S_1(\mathcal{C}) \cup S_2(\mathcal{C}) \subseteq \mathcal{C}.$$

The two steps above shows that

$$S_1(\mathcal{C}) \cup S_2(\mathcal{C}) = \mathcal{C}.$$

2. Recall T , T_0 , T_1 , T_2 , $T_3, \dots$, $T_n, \dots$ from Worksheet 2. Define the set $\mathcal{S}$, known as the *Sierpinski Triangle*, by

$$\mathcal{S} = \bigcap_{n=0}^{\infty} T_n.$$

- (a) Show that all the vertices of T belong to $\mathcal{S}$.

Solution: Notice that

$$S_1(P) = \frac{P+P}{2} = P, \quad S_2(Q) = \frac{Q+Q}{2} = Q \text{ and } S_3(R) = \frac{R+R}{2} = R.$$

$$T_{n+1} := S_1(T_n) \cup S_2(T_n) \cup S_3(T_n).$$

We know that $\{P, Q, R\} \in T_0$. Let $\{P, Q, R\} \in T_n$ for some $n \geq 0$. Then

$$P = S_1(P) \in S_1(T_n) \subset T_{n+1},$$

$$Q = S_2(Q) \in S_2(T_n) \subset T_{n+1} \text{ and}$$

$$R = S_3(R) \in S_3(T_n) \subset T_{n+1}.$$

By the principle of induction,

$$\{P, Q, R\} \in T_n \forall n \geq 0 \implies \{P, Q, R\} \in \mathcal{S}.$$

(b) Which of the following points lie in $\mathcal{S}$?

$$\begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3/4 \\ \sqrt{3}/4 \end{pmatrix}, \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix}$$

Solution: We will show that

$$\begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3/4 \\ \sqrt{3}/4 \end{pmatrix}, \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix} \in \mathcal{S}.$$

- i. $Q \in T_n \implies S_1(Q) = \frac{Q+P}{2} = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix} \in T_{n+1}.$
- ii. $R \in T_n \implies S_2(R) = \frac{Q+R}{2} = \begin{pmatrix} 3/4 \\ \sqrt{3}/4 \end{pmatrix} \in T_{n+1}.$
- iii. $R \in T_n \implies S_1(R) = \frac{R+P}{2} = \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix} \in T_{n+1}.$

We have shown in 2a that $\{P, Q, R\} \subset \mathcal{S} \implies \{P, Q, R\} \subset T_n \forall n \geq 0$. Thus

$$\begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3/4 \\ \sqrt{3}/4 \end{pmatrix}, \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix} \in T_n \forall n \geq 1.$$

It is easy to see that

$$\begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3/4 \\ \sqrt{3}/4 \end{pmatrix}, \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix} \in T_0.$$

The above steps shows that

$$\begin{aligned} & \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3/4 \\ \sqrt{3}/4 \end{pmatrix}, \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix} \in T_n \forall n \geq 0 \\ \implies & \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3/4 \\ \sqrt{3}/4 \end{pmatrix}, \begin{pmatrix} 1/4 \\ \sqrt{3}/4 \end{pmatrix} \in \mathcal{S}. \end{aligned}$$

(c) What is the total of the areas of all the triangles in T_0, T_1, T_2 ? What is it for T_n ?

Solution: We have shown in 2b that T_0 has $N_0 = 1$ equilateral triangle of area $A_0 = \frac{\sqrt{3}}{4}$. We have also shown in 3a that $N_{n+1} = 3N_n$ and $A_{n+1} = \frac{A_n}{4} \forall n \geq 0$. This recursion along with the initial values gives us the closed forms

$$N_n = 3^n \text{ and } A_n = \frac{\sqrt{3}}{4} \cdot \frac{1}{4^n} \forall n \geq 0.$$

Thus the total of the areas equals

$$N_n \cdot A_n = \frac{\sqrt{3}}{4} \cdot \left(\frac{3}{4}\right)^n \forall n \geq 0.$$

(d) Does the total of the areas converge as $n \rightarrow \infty$? If yes, what is the limit?

Solution: By 1e, $\left(\frac{3}{4}\right)^n$ converges to 0. Therefore,

$$A_n = \frac{\sqrt{3}}{4} \cdot \left(\frac{3}{4}\right)^n \rightarrow 0 \text{ as } n \rightarrow \infty.$$

(e) Show that $\mathcal{S} = S_1(\mathcal{S}) \cup S_2(\mathcal{S}) \cup S_3(\mathcal{S})$.

Solution: Let $x \in \mathcal{S} = \bigcap_{n=0}^{\infty} T_n$. Thus $x \in T_n \ \forall n \geq 0$. More specifically, $x \in T_1 = S_1(T_0) \cup S_2(T_0) \cup S_3(T_0)$.

- Case 1: $x \in S_1(T_0)$

$$S_1(\mathcal{S}) = \left\{ \frac{x+P}{2} \mid x \in \mathcal{S} \right\}.$$

Let $y = 2x - P$.

If $y \in \mathcal{S}$, then

$$x = \frac{y+P}{2} \in S_1(\mathcal{S}).$$

If $y \notin \mathcal{S}$, then

$$\exists n \in \mathbb{N} \text{ such that } y \notin T_n.$$

But since $x \in \mathcal{S}$, we have $x \in T_{n+1} = S_1(T_n) \cup S_2(T_n) \cup S_3(T_n)$. By assumption, $x \in S_1(T_n)$. Then $\exists a \in T_n$ such that

$$x = \frac{a+P}{2} \Rightarrow a = 2x - P = y,$$

contradicting $y \notin T_n$. Thus, $y \in \mathcal{S}$ and we are done.

- Case 2: $x \in S_2(T_0)$

$$S_2(\mathcal{S}) = \left\{ \frac{x+Q}{2} \mid x \in \mathcal{S} \right\}.$$

Let $y = 2x - Q$.

If $y \in \mathcal{S}$, then

$$\exists n \in \mathbb{N} \text{ such that } y \notin T_n.$$

If $y \notin \mathcal{S}$, then

$$\exists n \in \mathbb{N} \text{ such that } y \notin T_n.$$

But since $x \in \mathcal{S}$, we have $x \in T_{n+1} = S_1(T_n) \cup S_2(T_n) \cup S_3(T_n)$. By assumption, $x \in S_2(T_n)$. Then $\exists a \in T_n$ such that

$$x = \frac{a+Q}{2} \Rightarrow a = 2x - Q = y,$$

contradicting $y \notin T_n$. Thus, $y \in \mathcal{S}$ and we are done.

- Case 3: $x \in S_3(T_0)$

$$S_3(\mathcal{S}) = \left\{ \frac{x+R}{2} \mid x \in \mathcal{S} \right\}.$$

Let $y = 2x - R$.

If $y \in \mathcal{S}$, then

$$\exists n \in \mathbb{N} \text{ such that } y \notin T_n.$$

If $y \notin \mathcal{S}$, then

$$\exists n \in \mathbb{N} \text{ such that } y \notin T_n.$$

But since $x \in \mathcal{S}$, we have $x \in T_{n+1} = S_1(T_n) \cup S_2(T_n) \cup S_3(T_n)$. By assumption, $x \in S_3(T_n)$. Then $\exists a \in T_n$ such that

$$x = \frac{a+R}{2} \Rightarrow a = 2x - R = y,$$

contradicting $y \notin T_n$. Thus, $y \in \mathcal{S}$ and we are done.

Therefore,

$$\mathcal{S} \subseteq S_1(\mathcal{S}) \cup S_2(\mathcal{S}) \cup S_3(\mathcal{S}).$$

Now let $x \in S_1(\mathcal{S}) \cup S_2(\mathcal{S}) \cup S_3(\mathcal{S})$.

- Case 1: $x \in S_1(\mathcal{S})$. Then $x = \frac{y+P}{2}$ for some $y \in \mathcal{S}$. Now, $y \in T_n \Rightarrow x = \frac{y+P}{2} \in T_{n+1}$. Since $y \in \mathcal{S}$, $y \in T_n \forall n \geq 0$. Thus $x \in T_n \forall n \geq 1$. Since $T_0 \supseteq T_1$, $x \in T_n \forall n \geq 0$. Thus $x \in \mathcal{S}$.
- Case 2: $x \in S_2(\mathcal{S})$. Then $x = \frac{y+Q}{2}$ for some $y \in \mathcal{S}$. Now, $y \in T_n \Rightarrow x = \frac{y+Q}{2} \in T_{n+1}$. Since $y \in \mathcal{S}$, $y \in T_n \forall n \geq 0$. Thus $x \in T_n \forall n \geq 1$. Since $T_0 \supseteq T_1$, $x \in T_n \forall n \geq 0$. Thus $x \in \mathcal{S}$.
- Case 3: $x \in S_3(\mathcal{S})$. Then $x = \frac{y+R}{2}$ for some $y \in \mathcal{S}$. Now, $y \in T_n \Rightarrow x = \frac{y+R}{2} \in T_{n+1}$. Since $y \in \mathcal{S}$, $y \in T_n \forall n \geq 0$. Thus $x \in T_n \forall n \geq 1$. Since $T_0 \supseteq T_1$, $x \in T_n \forall n \geq 0$. Thus $x \in \mathcal{S}$.

Therefore,

$$S_1(\mathcal{S}) \cup S_2(\mathcal{S}) \cup S_3(\mathcal{S}) \subseteq \mathcal{S}.$$

Combining both inclusions, we conclude:

$$\boxed{\mathcal{S} = S_1(\mathcal{S}) \cup S_2(\mathcal{S}) \cup S_3(\mathcal{S})}.$$